

DSA0601 – DATA HANDLING AND VISUALIZATION

VISUALIZATION DESIGN EXERCISE – R PROGRAMMING

MAXIMUM MARKS: 100 | SLOT C | CO1 – AT3

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Q1. Integrated / Capstone Design [R]

Problem Statement: A financial analytics company has collected performance data for twenty publicly listed companies. Develop an integrated visualization that communicates Revenue, Profit Margin, Market Capitalization, Employee Strength, and Industry Sector in one coherent analytical view.

Company	Sector	Revenue	Profit Margin	Employees	Market Cap
C1	IT	420	8.2	1500	8500
C2	Banking	580	15.6	3200	12000
C3	Healthcare	720	11.4	2100	9800
C4	Retail	850	9.3	4200	7600
C5	Energy	980	18.5	2800	14500
C6	IT	1150	20.4	5100	18200
C7	Banking	1320	17.8	6800	21500
C8	Healthcare	1480	14.2	3900	16800
C9	Retail	1650	10.5	7200	13200
C10	Energy	1820	22.1	4500	24500
C11	IT	2050	24.5	8100	28500
C12	Banking	2280	19.8	9500	32500
C13	Healthcare	2550	16.7	6200	24800
C14	Retail	2840	13.4	10300	19200
C15	Energy	3150	26.2	7200	38200
C16	IT	3480	28.1	11800	42500
C17	Banking	3860	21.5	13200	46800
C18	Healthcare	4250	18.3	8600	35600
C19	Retail	4680	15.2	14500	29800
C20	Energy	5200	30.4	16000	52000

What the visualization should communicate

- Revenue should be mapped to the horizontal position because position is easy to compare accurately.
- Profit Margin should be mapped to the vertical position so companies can be compared in terms of profitability.
- Market Capitalization can be mapped to point size because it is a quantitative variable.
- Sector can be mapped to color because it is categorical.
- Employee Strength may be represented through an additional aesthetic only if it remains readable; otherwise, explain why it is better handled in a companion plot.

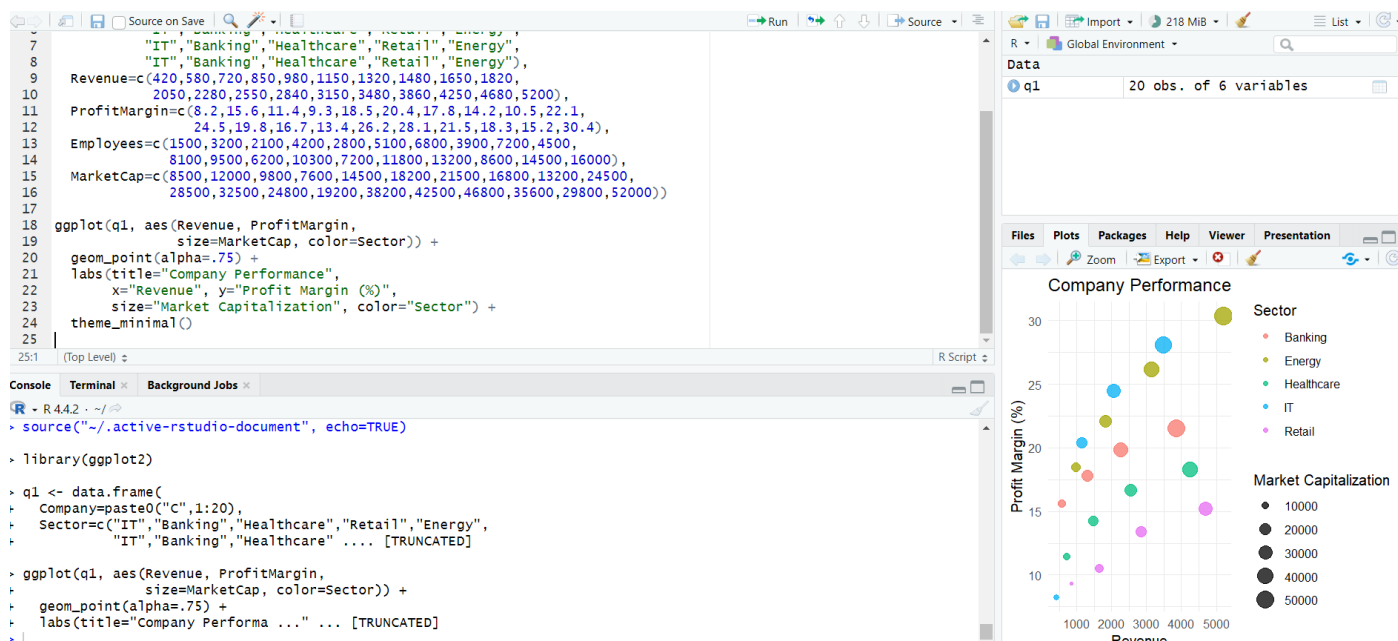
R Code – Q1

```
library(ggplot2)

q1 <- data.frame(
  Company=paste0("C",1:20),
  Sector=c("IT","Banking","Healthcare","Retail","Energy",
           "IT","Banking","Healthcare","Retail","Energy",
           "IT","Banking","Healthcare","Retail","Energy",
           "IT","Banking","Healthcare","Retail","Energy"),
  Revenue=c(420,580,720,850,980,1150,1320,1480,1650,1820,
            2050,2280,2550,2840,3150,3480,3860,4250,4680,5200),
  ProfitMargin=c(8.2,15.6,11.4,9.3,18.5,20.4,17.8,14.2,10.5,22.1,
                24.5,19.8,16.7,13.4,26.2,28.1,21.5,18.3,15.2,30.4),
  Employees=c(1500,3200,2100,4200,2800,5100,6800,3900,7200,4500,
              8100,9500,6200,10300,7200,11800,13200,8600,14500,16000),
  MarketCap=c(8500,12000,9800,7600,14500,18200,21500,16800,13200,24500,
              28500,32500,24800,19200,38200,42500,46800,35600,29800,52000))

ggplot(q1, aes(Revenue, ProfitMargin,
               size=MarketCap, color=Sector)) +
```

```
geom_point(alpha=.75) +
labs(title="Company Performance",
      x="Revenue", y="Profit Margin (%)",
      size="Market Capitalization", color="Sector") +
theme_minimal()
```



Explanation / Interpretation

This is an integrated bubble scatter plot. Each point represents one company. Horizontal and vertical positions provide the most precise quantitative comparisons, while bubble size communicates market capitalization and color separates industry sectors. The plot can reveal clusters, unusual companies, and relationships between revenue and profitability. Employee strength is retained in the dataset and can be examined using a second supporting plot if adding another aesthetic would make the main plot confusing.

Component	Marks
Correct data entry and preparation	5
Appropriate aesthetic mappings	8
Correct ggplot2 implementation	7
Title, labels, legend and scale	4
Interpretation and design justification	6
Total Q1	30

Q2. Multi-Aesthetic Design [R]

Problem Statement: A multinational retail company has collected annual sales information from branches in different regions. Create a visualization that simultaneously represents Sales, Profit, Customers, Product Category, and Region.

Branch	Region	Category	Sales	Profit	Customers
B1	North	Electronics	450000	45000	1200
B2	South	Furniture	620000	95000	2500
B3	East	Clothing	180000	-12000	980
B4	West	Food	920000	185000	4200
B5	North	Electronics	350000	25000	1800
B6	South	Furniture	720000	120000	3000
B7	East	Clothing	250000	18000	1250
B8	West	Food	1080000	225000	5200
B9	North	Electronics	560000	72000	2100
B10	South	Furniture	670000	98000	3300
B11	East	Clothing	195000	-8000	1100
B12	West	Food	980000	205000	4700
B13	North	Electronics	430000	51000	1750
B14	South	Furniture	760000	132000	3600

B15	East	Clothing	280000	24000	1450
B16	West	Food	1150000	250000	5600
B17	North	Electronics	510000	69000	2200
B18	South	Furniture	690000	115000	3900
B19	East	Clothing	220000	15000	1600
B20	West	Food	1250000	285000	6100

Required aesthetic mapping

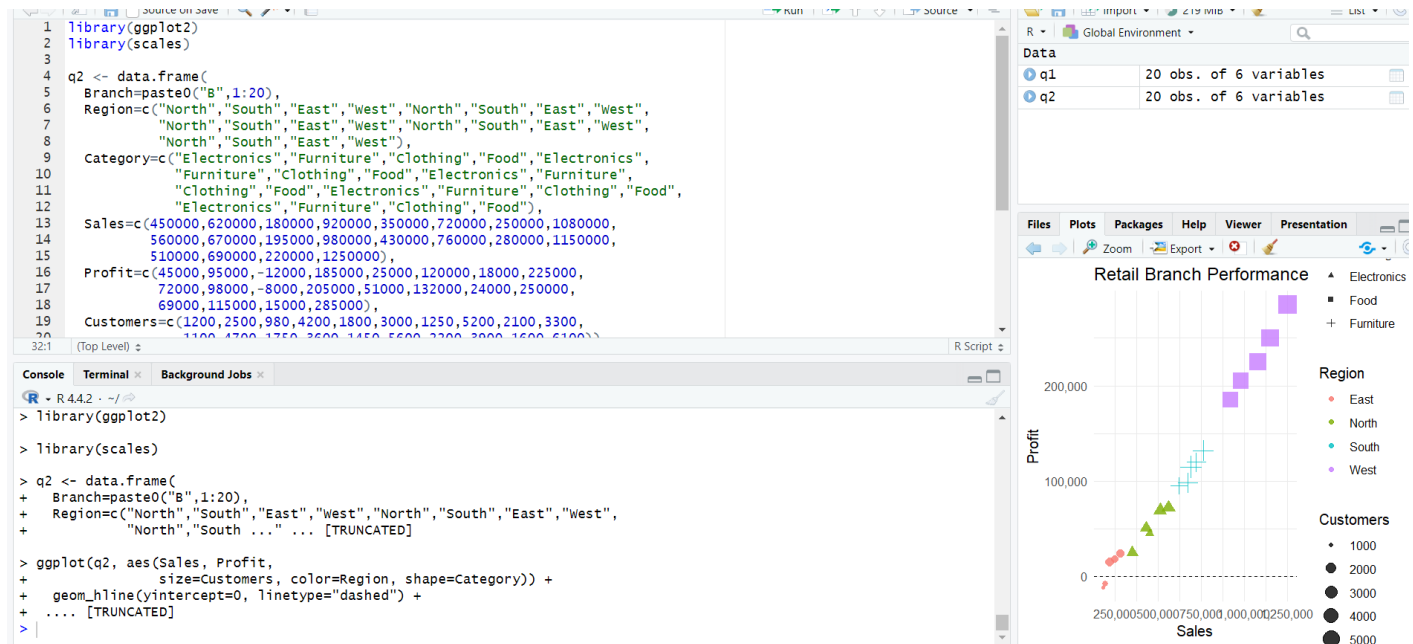
- Sales → x-position; Profit → y-position.
- Customers → point size.
- Region → color.
- Product Category → shape.
- Keep negative profit values visible; branches with losses must appear below the zero-profit line.

R Code – Q2

```
library(ggplot2)
library(scales)

q2 <- data.frame(
  Branch=paste0("B",1:20),
  Region=c("North", "South", "East", "West", "North", "South", "East", "West",
           "North", "South", "East", "West", "North", "South", "East", "West",
           "North", "South", "East", "West"),
  Category=c("Electronics", "Furniture", "Clothing", "Food", "Electronics",
            "Furniture", "Clothing", "Food", "Electronics", "Furniture",
            "Clothing", "Food", "Electronics", "Furniture", "Clothing", "Food",
            "Electronics", "Furniture", "Clothing", "Food"),
  Sales=c(450000, 620000, 180000, 920000, 350000, 720000, 250000, 1080000,
          560000, 670000, 195000, 980000, 430000, 760000, 280000, 1150000,
          510000, 690000, 220000, 1250000),
  Profit=c(45000, 95000, -12000, 185000, 25000, 120000, 18000, 225000,
          72000, 98000, -8000, 205000, 51000, 132000, 24000, 250000,
          69000, 115000, 15000, 285000),
  Customers=c(1200, 2500, 980, 4200, 1800, 3000, 1250, 5200, 2100, 3300,
             1100, 4700, 1750, 3600, 1450, 5600, 2200, 3900, 1600, 6100))

ggplot(q2, aes(Sales, Profit,
               size=Customers, color=Region, shape=Category)) +
  geom_hline(yintercept=0, linetype="dashed") +
  geom_point(alpha=.8) +
  scale_x_continuous(labels=comma) +
  scale_y_continuous(labels=comma) +
  labs(title="Retail Branch Performance",
       x="Sales", y="Profit",
       size="Customers", color="Region", shape="Category") +
  theme_minimal()
```



Explanation / Interpretation

The plot uses multiple aesthetics without making the reader decode the same variable in several ways. Position carries the two main financial measures, size represents customer volume, color identifies region, and shape identifies product category. The zero-profit reference line makes loss-making branches immediately identifiable.

Component	Marks
Correct data preparation	4
Multi-aesthetic mapping	7
Correct R/ggplot2 code	6
Readability and labels	3
Interpretation and justification	5
Total Q2	25

Q3. Nonlinear Axis for Skewed Data [R] — 20 Marks

Problem Statement: A software company has recorded project statistics for twenty development teams. Lines of Code is highly right-skewed. Create a visualization using a nonlinear axis transformation and explain why it improves interpretation compared with a linear scale.

Team	Lines of Code	Bugs	Developers	Complexity
T1	12000	35	5	2.1
T2	18000	52	7	2.4
T3	25000	60	9	2.7
T4	35000	72	10	3.0
T5	48000	88	12	3.2
T6	62000	110	14	3.5
T7	81000	125	16	3.8
T8	96000	142	18	4.1
T9	120000	165	20	4.5
T10	150000	188	22	4.8
T11	185000	205	25	5.1
T12	240000	240	28	5.4
T13	310000	280	30	5.8
T14	390000	320	34	6.2
T15	480000	365	38	6.6
T16	620000	420	42	7.1
T17	810000	490	46	7.6
T18	1050000	560	50	8.0
T19	1350000	640	55	8.5
T20	1800000	720	60	9.0

Why a nonlinear axis is necessary

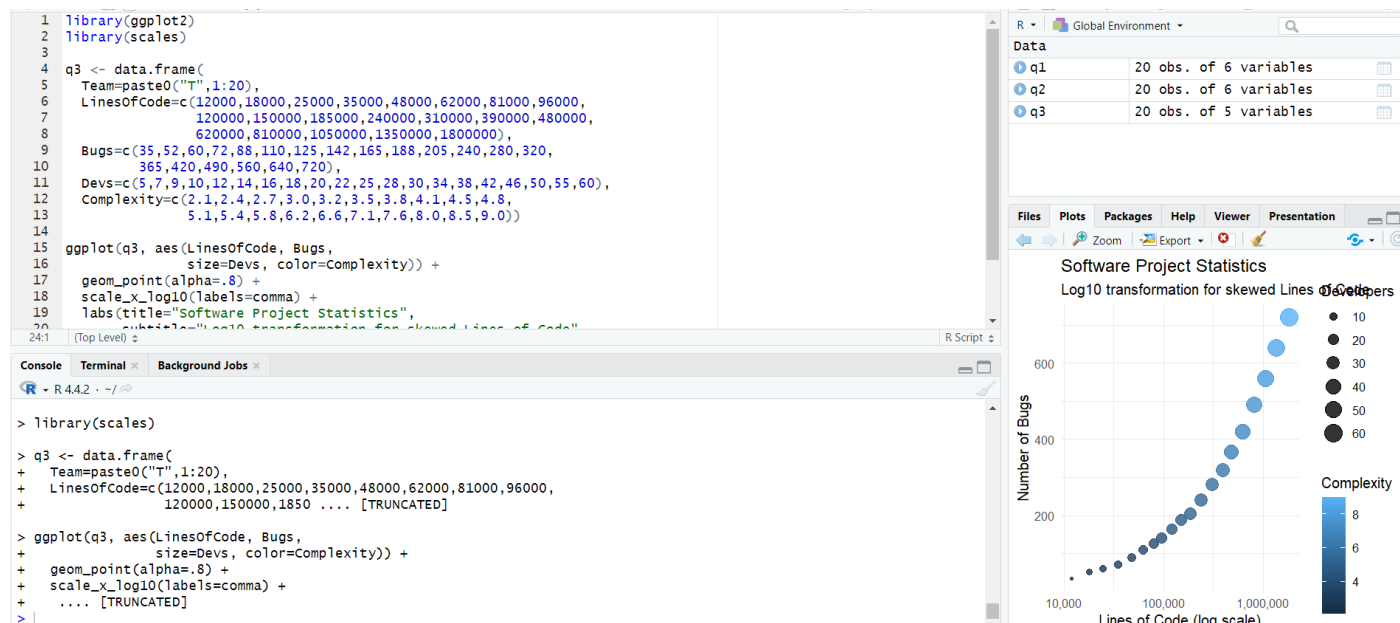
Lines of Code ranges from 12,000 to 1,800,000. On a normal linear axis, the smaller projects would be compressed close to the origin because the largest values are much bigger. A logarithmic transformation spreads the values across the plotting area while preserving their ordering.

R Code – Q3

```
library(ggplot2)
library(scales)

q3 <- data.frame(
  Team=paste0("T",1:20),
  LinesOfCode=c(12000,18000,25000,35000,48000,62000,81000,96000,
    120000,150000,185000,240000,310000,390000,480000,
    620000,810000,1050000,1350000,1800000),
  Bugs=c(35,52,60,72,88,110,125,142,165,188,205,240,280,320,
    365,420,490,560,640,720),
  Devs=c(5,7,9,10,12,14,16,18,20,22,25,28,30,34,38,42,46,50,55,60),
  Complexity=c(2.1,2.4,2.7,3.0,3.2,3.5,3.8,4.1,4.5,4.8,
    5.1,5.4,5.8,6.2,6.6,7.1,7.6,8.0,8.5,9.0))

ggplot(q3, aes(LinesOfCode, Bugs,
  size=Devs, color=Complexity)) +
  geom_point(alpha=.8) +
  scale_x_log10(labels=comma) +
  labs(title="Software Project Statistics",
    subtitle="Log10 transformation for skewed Lines of Code",
    x="Lines of Code (log scale)", y="Number of Bugs",
    size="Developers", color="Complexity") +
  theme_minimal()
```



Explanation / Interpretation

A log10 x-axis is appropriate because the independent variable spans more than two orders of magnitude. The transformation does not change the original data; it changes the visual scale. This prevents large projects from dominating the horizontal space and makes smaller ones more visible.

Q4. Cartesian vs. Polar Coordinates [R]

Problem Statement: The dataset contains monthly electricity generation from Coal, Gas, Solar, Wind, and Hydro sources. Produce both Cartesian and Polar-coordinate visualizations and explain how the coordinate system changes the analytical reading of the data.

Month	Coal	Gas	Solar	Wind	Hydro
Jan	620	310	45	80	140
Feb	580	295	60	95	150
Mar	590	305	90	120	160
Apr	605	320	130	150	170

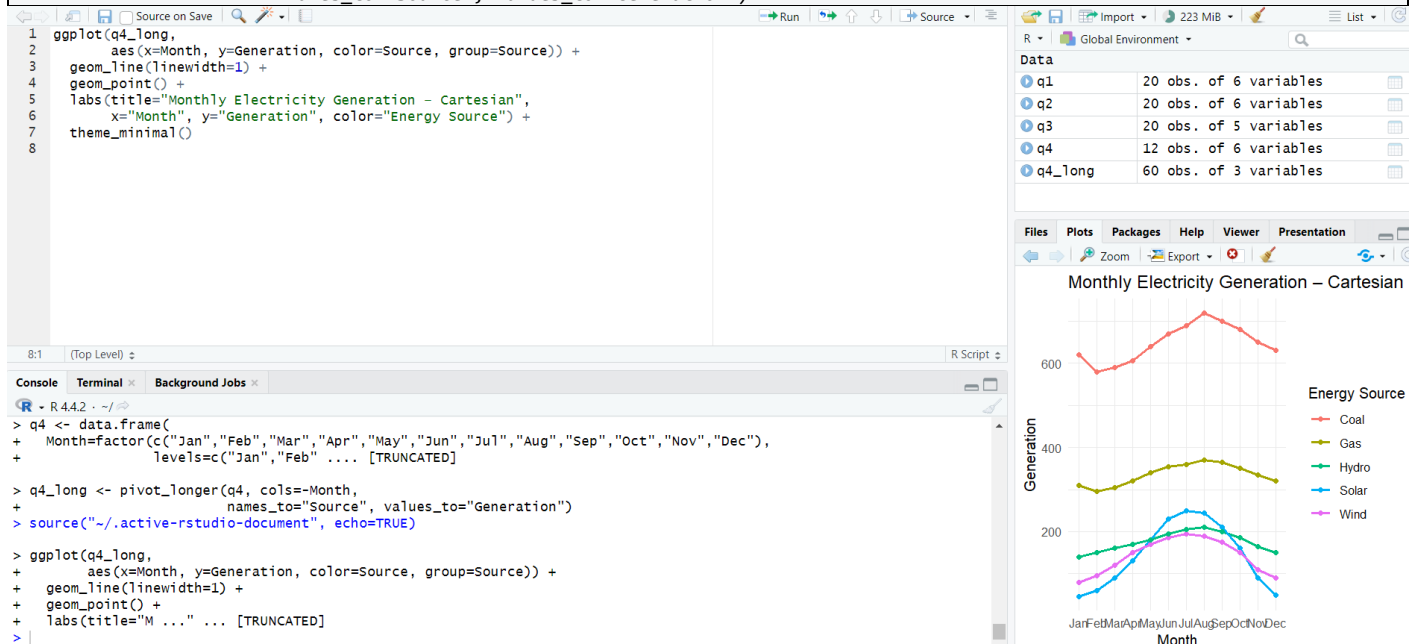
May	640	340	180	170	180
Jun	670	355	230	185	195
Jul	690	360	250	195	205
Aug	720	370	245	190	210
Sep	700	365	210	175	200
Oct	680	350	160	150	185
Nov	650	335	90	110	165
Dec	630	320	50	90	150

R Code – Q4 Data Preparation

```
library(ggplot2)
library(tidyr)

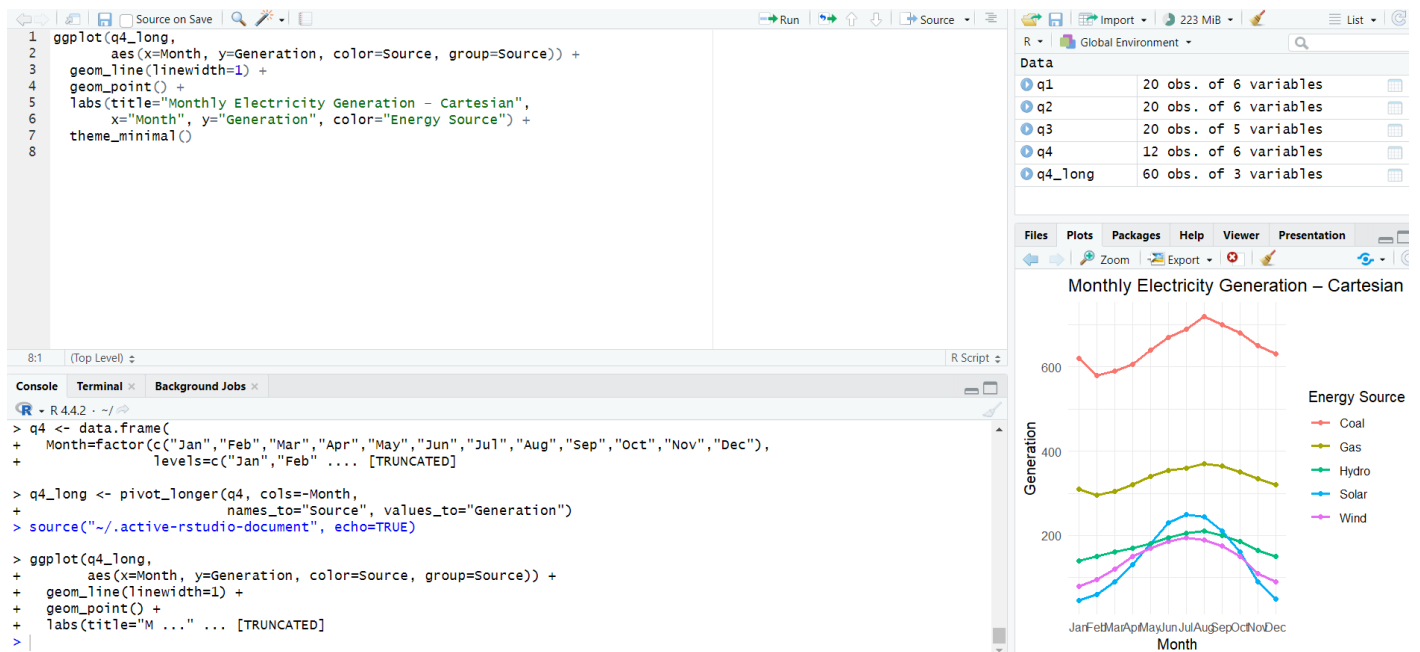
q4 <- data.frame(
  Month=factor(c("Jan", "Feb", "Mar", "Apr", "May", "Jun", "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"),
    levels=c("Jan", "Feb", "Mar", "Apr", "May", "Jun", "Jul", "Aug", "Sep", "Oct", "Nov", "Dec")),
  Coal=c(620,580,590,605,640,670,690,720,700,680,650,630),
  Gas=c(310,295,305,320,340,355,360,370,365,350,335,320),
  Solar=c(45,60,90,130,180,230,250,245,210,160,90,50),
  Wind=c(80,95,120,150,170,185,195,190,175,150,110,90),
  Hydro=c(140,150,160,170,180,195,205,210,200,185,165,150))

q4_long <- pivot_longer(q4, cols=-Month,
  names_to="Source", values_to="Generation")
```

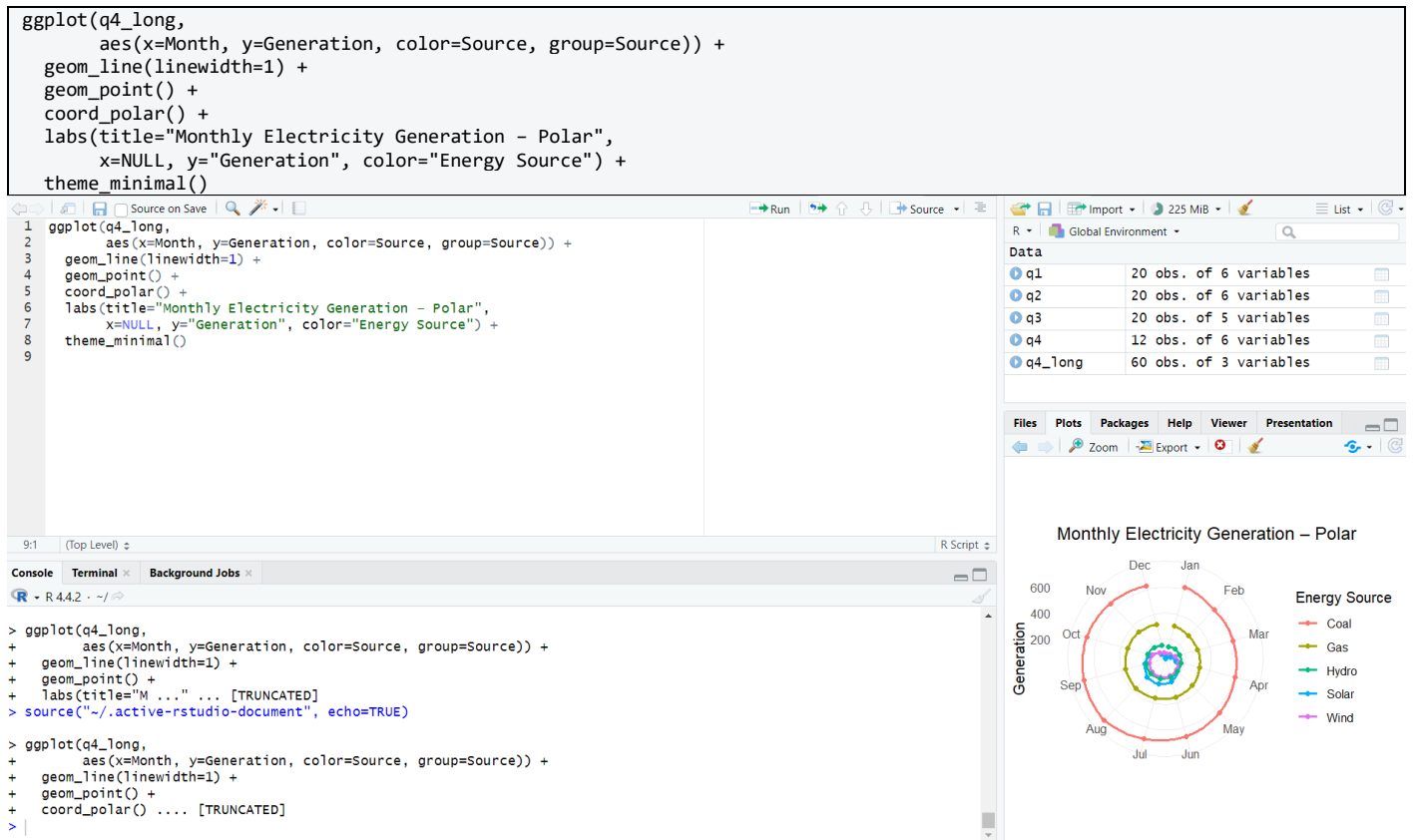


R Code – Q4A: Cartesian Coordinates

```
ggplot(q4_long,
  aes(x=Month, y=Generation, color=Source, group=Source)) +
  geom_line(linewidth=1) +
  geom_point() +
  labs(title="Monthly Electricity Generation – Cartesian",
    x="Month", y="Generation", color="Energy Source") +
  theme_minimal()
```



R Code – Q4B: Polar Coordinates



Explanation / Interpretation

Cartesian coordinates place months on a straight horizontal axis. This makes exact month-to-month comparisons and differences between sources easier to read. Polar coordinates wrap the months around a circle, which can emphasize the cyclical nature of the annual sequence. However, distances and angles can make precise quantitative comparisons harder. Therefore, the two plots serve different analytical purposes rather than being interchangeable.